

NYU, Tandon School of Engineering

Bridge to CS

Homework #8

Submission Instructions

1. For this assignment, you should turn in 6 `.cpp` files:
 - Each question requires one file.
 - Name your files as follows:
 - Q1: `hw8.q1.cpp`
 - Q2: `hw8.q2.cpp`
 - Q3: `hw8.q3.cpp`
 - Q4: `hw8.q4.cpp`
 - Q5: `hw8.q5.cpp`
 - Q6: `hw8.q6.cpp`
 - You should test your code by using a main function. However, **do not include any main functions in your final submissions.**
 - Some of these problems have asymptotic runtime requirements (Θ). Please complete the C10: Introduction to Algorithm Analysis Module before attempting these problems.
 2. Pay special attention to the style of your code. Indent your code correctly, choose meaningful names for your variables, define constants where needed, choose the most appropriate control flow statements, break down your solutions by defining functions, etc.
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Question 1: Minimum in Array

Implement the following functions:

1. `int minInArray(int arr[], int arrSize)`

This function is given `arr`, an array of integers, and its logical size, `arrSize`. When called, it returns the minimum value in `arr`.

2. `void findMinIndices(int arr[], int arrSize, int minVal)`

This function is given an array, its size, and the minimum value. It prints all the indices where the minimum value appears, separated by spaces.

For example, given the array `{14, 5, 12, 5, 6, 14, 5, 12, 14, 12, 14, 6, 8, 7, 5, 136, 9, 2189, 10, 6}`:

- `minInArray(arr, 20)` returns 5
- `findMinIndices(arr, 20, 5)` prints:

```
1 3 6 14
```

Note: The indices should be printed on a single line, separated by spaces.

Question 2: Palindrome

A **palindrome** is a word which reads the same backward or forward. For example, *noon*, *civic*, *radar*, *level*, *rotor*, *kayak*, *reviver*, *racecar*, *redder*, *madam*, and *refer* are all palindromes.

Implement the following function:

```
bool isPalindrome(string str)
```

This function is given a string `str` containing a word, and returns `true` if and only if `str` is a palindrome.

Examples:

- `isPalindrome("level")` returns `true`
- `isPalindrome("racecar")` returns `true`
- `isPalindrome("hello")` returns `false`
- `isPalindrome("a")` returns `true`

Question 3: Array Manipulations

Implement the following functions:

a) `void reverseArray(int arr[], int arrSize)`

Takes `arr`, an array of integers, and its size, `arrSize`. When called, it reorders the elements of the array to appear in reverse order.

For example, if `arr` is [1, 2, 3, 4], after calling `reverseArray`, `arr` will be [4, 3, 2, 1].

b) `void removeOdd(int arr[], int& arrSize)`

Takes `arr`, an array of integers, and its size, `arrSize` (passed by reference). When called, the function alters `arr` so that only the even numbers remain, in their original relative order. Additionally, the function updates `arrSize` to contain the new logical size.

For example, if `arr` is [1, 2, 3, 4], after calling `removeOdd`, `arr` will be [2, 4], and `arrSize` will be 2.

c) `void splitParity(int arr[], int arrSize)`

Takes `arr`, an array of integers, and its size, `arrSize`. When called, the function changes the order of numbers in `arr` so that all the odd numbers appear first, and all the even numbers appear last. The inner order of odd and even numbers does not matter.

For example, if `arr` is [1, 2, 3, 4], after calling `splitParity`, `arr` could be [3, 1, 2, 4] or [1, 3, 4, 2], etc.

Implementation Requirements:

1. In all three functions, you are **not allowed** to use an auxiliary array (a temporary local array).
2. Pay attention to the running time of your functions. An efficient implementation runs in linear time $\Theta(\text{arrSize})$.

Expected output when testing with sample arrays:

```
-3 12 14 2 9
12 6 14
1 3 6 4 12
```

(Note: The output for `splitParity` may vary as long as odds come before evens.)

Question 4: PIN Authentication

Traditional password entry schemes are susceptible to “shoulder surfing” in which an attacker watches an unsuspecting user enter their password or PIN number. One way to combat this is with a randomized challenge-response system.

Consider a scheme where the password consists of a five-digit PIN number (00000 to 99999). Each digit is assigned a random number that is 1, 2, or 3. The user enters the random numbers that correspond to their PIN instead of their actual PIN numbers.

For example, with PIN 12345 and mapping:

```
PIN: 0 1 2 3 4 5 6 7 8 9
NUM: 3 2 3 1 1 3 2 2 1 3
```

The user would enter 23113 instead of 12345.

Implement the following functions:

1. `void generateMapping(int mapping[])`
Fills the `mapping` array (size 10) with random numbers 1, 2, or 3 for each digit 0-9.
2. `void printMapping(int mapping[])`
Prints the PIN-to-NUM mapping in the format shown below.
3. `bool checkPIN(int actualPIN, int enteredCode, int mapping[])`
Returns `true` if the `enteredCode` correctly corresponds to the `actualPIN` given the `mapping`.

Example execution (assuming actual PIN is 12345):

```
Please enter your PIN according to the following mapping:
PIN: 0 1 2 3 4 5 6 7 8 9
NUM: 3 2 3 1 1 3 2 2 1 3
23113
Your PIN is correct
```

Another execution (same PIN, different mapping):

```
Please enter your PIN according to the following mapping:
PIN: 0 1 2 3 4 5 6 7 8 9
NUM: 1 1 2 3 1 2 2 3 3 3
23113
Your PIN is not correct
```

Question 5: Name Formatting

Write a function that reads a person's name in the format: first name, then middle name or initial, and then last name. The function outputs the name in the format: `Last_Name, First_Name Middle_Initial`.

Implement the following function:

```
void formatName(string first, string middle, string last)
```

This function takes the three parts of a name and prints the formatted output.

Examples:

- Input: "Mary", "Average", "User"

Output:

```
User, Mary A.
```

- Input: "Mary", "A.", "User"

Output:

```
User, Mary A.
```

- Input: "John", "William", "Smith"

Output:

```
Smith, John W.
```

Note: The function should place a period after the middle initial even if the input did not contain one.

Question 6: Hide Standalone Numbers

Write a function that takes a line of text and returns the line with all the digits in **standalone integer numbers** replaced with 'x'.

Implement the following function:

```
string hideNumbers(string text)
```

This function takes a string and returns a new string with standalone numbers replaced.

Example:

- Input: "My userID is john17 and my 4 digit pin is 1234 which is secret"

Output:

```
My userID is john17 and my x digit pin is xxxx which is secret
```

Notes:

1. If a digit is part of a word (like john17), the digit is **NOT** changed to 'x'.
2. A standalone number is a sequence of digits surrounded by spaces (or at the beginning/end of the string).
3. You may assume that the text contains only letters (upper or lower case), digits, and spaces.

More examples:

- `hideNumbers("I have 2 cats")` returns `"I have x cats"`
- `hideNumbers("Room 101 has 25 seats")` returns `"Room xxx has xx seats"`
- `hideNumbers("abc123def")` returns `"abc123def"` (not standalone)
- `hideNumbers("123")` returns `"xxx"`